



Data Management Perspectives on Business Process Management

(Tutorial Overview)

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ABSTRACT

Traditional approaches to Business Process Management (BPM) focus primarily on the process aspects, and treat the persistent data accessed and manipulated by the business processes as second-class citizens. A recent approach to BPM, based on “business artifacts”, is centered on a modeling framework that places data and process on an equal footing. The approach has been shown useful in various application domains, and one variant of business artifacts forms the basis of the emerging OMG Case Management Model and Notation (CMMN) standard. Research results have been developed around conceptual models, enterprise interoperability, business intelligence, and verification. This data-centric approach has the potential to provide the basis for a new generation of BPM technology in support of diverse application, and fueled by the insights into abstraction and data management that have been the hallmark of database research since the 70’s.

Categories and Subject Descriptors

H.4.1 [Information Systems Applications]: Office Automation—*workflow management*; K.6.3 [Management of Computing and Information Systems]: Software Management

General Terms

Management

Keywords

BPM, business processes, business artifacts, process modeling

1. INTRODUCTION

Enterprise Systems are large-scale software applications that support data management, business processes, information flows, reporting, and data analytics in a complex organization. An enterprise system is typically based on a combination of one or more database management systems, one or more business process management systems, one or more reporting systems to support busi-

ness intelligence, along with other systems (e.g., web access, authentication). Two key components of an enterprise system are management of data and management of business processes.

A *business process* (or *BP*) generally refers to an assembly of activities (tasks) possibly performed by humans to achieve one or more business goals. BPs are ubiquitous and occur in all sectors: government agencies (e.g., approval of licenses or permits), universities and institutions (e.g., application review), funding agencies and conference organizations (e.g., paper review), hospitals (e.g., out-patient care), financial management (e.g., travel reimbursements), human resource management (e.g., hiring), and of course, business (e.g., order fulfillments). While the number of BP schemas (or models) may range from hundreds (e.g. 500 in REA, a real estate administration office in a Chinese city [37]) to hundreds of thousands (e.g. 200K in CNR [16]), the number of process executions (or cases) per year is often much higher (e.g. 300K for REA). Management of BP schemas and BP executions is very challenging and is costing a substantial amount of money for organizations world wide. An important goal of the “Big Data” [20] analysis is to discover implicit knowledge that can improve BP schemas and make them more efficient and effective, and consume less resources. However, the unfortunate reality is that the current development and practice of BP management (or BPM) is not well guided by suitable methodologies and techniques, and consequently, almost everything beyond running BPs and keeping records becomes a high risk research project. For example, it can become very costly to set up a new BP (including integration with already existing processes, data management systems, etc.), and it may be impossible to significantly modify BP schemas (e.g., to accommodate business mergers or changing business conditions).

At a fundamental level, BPM provides structured approaches for systematic management of collaborations of individuals in achieving business (or enterprise) goals. Specifically, it includes management and support for design, execution, reasoning/analysis, and modification/evolution of BP schemas, and also management of all needed resources (including human performers) to ensure integrity and completion of execution of BPs, handling of exceptional cases, compliance to laws and regulations. One aspect of BPM was studied in the database community more than 20 years ago: the work on workflow systems narrowly focused on chaining together data access activities (that occur in a BP execution) to extend the notion of a database transaction. The work quickly revealed the difficulty of extending the ACID properties to “workflow transactions”.

In the last decade, a significant amount of effort has been devoted to modeling of BPs by focusing on tasks/activities and control flow among them. Among the research issues studied are verification of BP schemas, properties of BP schemas concerning structural well-formedness, variability, similarity, etc. In the development commu-

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nity, standards such as BPMN (Business Process Model and Notations), BPEL (Business Process Execution Language), and SBVR (Semantics of Business Vocabulary and Rules), etc. are formulated with the intent to be used in BPM development. However, a key omission in these languages is that their models do not include data as first-class citizens; this omission limits applicability of research results and leaves many important implementation decisions to ad hoc solutions.

BPs are intimately related to *persistent* data. BP executions usually generate a significant amount of data, much of which is in the scope of Big Data. For example, NTDB (National Trauma Data Bank) has been collecting more than a half million records for trauma patients each year since 2006, and large retailers are collecting data on thousands of transactions and user click streams daily. Obtaining useful business insights about the overall functioning of an enterprise system is challenging because of the broad heterogeneity across the underlying systems (multiple BPM systems, multiple databases, and multiple reporting systems). Business users need end-to-end, cross-functional, and integrated views that focus on tangible business milestones and outcomes, rather than details about how many times a given node of an activity flow has been visited or the history of how a database value has changed over time. Fundamental to the challenge of obtaining cross-functional views is the mismatch between how business processes are typically specified how the data is organized.

Modeling BPs with data is an emerging research topic with the potential to make a profound impact on the BP applications community. For example, the 2009 NSF Workshop on Data-Centric Workflows [15] enumerated several challenges for BPM and other forms of transactional workflow, and pointed to approaches that give data high prominence as holding the promise to overcome those challenges. More specifically, the workshop report notes that “the widely used workflow technologies do not provide adequate support for a variety of essential functionalities”, including low-risk design and deployment of BPs, automated tools to support BP interoperability, and tools for enabling BP schema re-use. The report goes on to say that “a key inhibitor in overcoming these challenges is the lack of intuitively clear ways for combining the various aspects of a workflow”, including “processes, data, people and automated agents, rules and incorporation of legacy applications and external services”. It then states that emerging approaches to BP modeling that place data and process on an equal footing holds “the promise of enabling substantial progress towards resolving many of today’s challenges in” BP and transactional workflow, precisely because this approach provides a unified, coherent basis around which the other aspects of BP can be layered. As indicated in the report, the task of fully understanding and leveraging this new approach to BP will require research along several dimensions, including conceptual models, design methodologies, reasoning about BP models and executions, systems issues, analytics on families of BP executions, and support for BP interoperability.

This tutorial will introduce the audience to three significant ways that the data management perspective is being applied to better understand business processes in enterprise systems. The early focus of database research around BPM was in the area of Business Intelligence (BI), that is, reporting and analytics focused on the data produced or impacted by business processes. A second focus has been to study both BP schemas and BP enactments (i.e., single executions of a process) as data objects in their own right. This work includes approaches to query both schemas and enactments, and also to perform process mining, that is, inferring BP schemas from a family of BP enactments. The third, and arguably most important, focus stems from a data-centric approach to business process,

an approach based on combining process and data at a fundamental level. Specifically, in the “business artifact” approach, introduced in 2003 [24, 17], the processes and operations of a business are specified using a family of “business artifacts”; these correspond to key conceptual entities that progress through the business, and explicitly model both the data and the lifecycle (process) aspects of those entities. This approach is also taken in the area of Case Management, which has emerged from domain-specific usage (primarily in social work, healthcare, and legal) to general-purpose application. This combination of data and process, at a fundamental level, leads to a simplification of several classical BPM challenges, including BI, BP integration and interoperability, and holds the promise of simplifying BPM variations and evolution. The approach also provides new ways to think about data. For example, it has given rise to the development of techniques for formal verification of systems that combine data with process. We believe that as more database researchers become aware of this data-centric approach to BPM there will be new classes of techniques and results that address open challenges in the BPM community. A central goal of the tutorial is to expose the larger database community to this data-centric approach to BPM, because it can provide the basis for some very stimulating research at the intersection between data management and business process management (and perhaps more generic kinds of process) over the coming years.

Specific topics covered in this tutorial are briefly discussed in Sections 2-5. In Section 2 a survey of BP approaches for incorporating data is presented, including the business artifact approach. Section 3 considers extensions of the artifact-centric approach, and Section 4 highlights the advantages of data-centric BP in the areas of Business Intelligence and process mining. Section 5 describes research on verification for data-centric BP.

2. DATA IN BUSINESS PROCESS MODELS

Data plays a critical role in enterprise systems. The first evidence is that data, together with its associated semantics, is a fundamental piece in formulating the semantics for BPs, i.e., what a BP actually does and how its actions are related to the environment/context. Consequently, data faithfully records the progress of individual BP executions (instances), including execution status, resource usage and status, and correlations with other BP instances. Further, executing a BP generates additional data for a variety of reasons, such as monitoring for performance or business concerns, auditing, compliance checking, etc. Finally, even BP schemas and enactments can be viewed as data so that they can be managed, queried, mined for (components of) process schemas, and analyzed.

If we narrowly focus on the aspect of BP schema design, there are four classes of data that are involved. Here we emphasize that by data, we mean *persistent data* properly managed in enterprise database systems. The first class is *business data* that are essential for the business logic (e.g. shipping address for a purchase BP). The second class of data holds the *status* of BP execution (e.g. shipping request has been made), this includes both conceptual data (meaningful to the user/performer of BP) and implementation specific data (needed for carrying out, e.g., database operations correctly). The third class captures *resource usage and status* for BP executions (e.g. cargo space reserved for the shipment in the morning delivery truck). And finally, the fourth class describes *correlations* between related BP execution instances (e.g. the purchase BP instance spawned three fulfillment BP instances).

In the literature, all BP modeling languages have the ability to specify tasks/activities and their process flow, but their ability to represent, use, and manipulate data involved vary. Roughly, process modeling languages can also roughly divided into four groups.

Data agnostic models essentially ignore three classes of data but focus on activities possibly with arrangement of activities using swimlanes and pools; execution status data is only partially available and implicit through the execution semantics. Petri nets, workflow nets, UML activity diagrams, and BPMN are typical examples in this group. *Data aware models* use variables to represent business data and thus are capable of specifying detailed computation logic for BPs. BPEL and YAWL [34] are representatives. While these modeling languages can be computationally expressive, they generally lack the explicit notion of (persistent) data representations (and needed abstractions) in the enterprise databases, which are important in managing BPs including schemas and executions. The third group includes *storage-aware models* (e.g., jBPM, UML including both class and activity diagrams) that support the concept of persistent data stores and modeling of data. However, mappings between persistent data and BP data are either not specified (UML) or specified only in the implementation level through SQL expressions (jBPM). The last group of BP models is best represented by *artifact-centric models*. Unlike the models in the previous groups the artifact-centric modeling approach starts from modeling data as a fundamental ingredient and specifies a BP around the evolution of (persistent) data entities in the form of lifecycle, in contrast to simply adding data as variables on top of activity based models, or including a logically detached data management component.

A central notion in the artifact-centric modeling approach is the notion of a *business artifact*, which is used to capture the key conceptual objects that evolve as they move through an enterprise. There are two essential components in the specification of a class of business artifacts: (i) a data schema (or information model) for holding information about the artifact as it moves from creation, through the process, and in some cases, to archival storage, and (ii) the lifecycle schema which describes how and when tasks (activities, or services) might be invoked on the artifacts as they move through the process. A prototypical example of a business artifact is the notion of “air courier package delivery”, whose data schema can hold information about a package including sender, receiver, the steps occurring in transport, and the billing activity, and whose lifecycle would specify the possible ways that the delivery service and billing might be carried out. Indeed, the typical package tracking information provided by commercial delivery services can be understood as providing a subset or “view” of the data value associated with the delivery artifact as it progresses through the courier’s BP, along with an abstracted view of the lifecycle, shown as the likely steps that will lead to completion of the enactment. Application systems based on business artifacts typically involve several distinct artifact types; communication and synchronization between related artifact instances must be supported.

Since its introduction in 2003, several variants of the business artifact approach have been studied in the literature and used in application. The variations focus mainly on how the lifecycles are specified. The first variation to be studied uses some form of finite state machines for the lifecycles; this has been applied in many areas, including integration of BPs, both horizontally (e.g., across silos in an enterprise) [4] and vertically (e.g., providing a unified view of similar BPs being performed in different regions) [7]. Some research on “business objects” adopts essentially the same approach as the finite-state-machine based artifacts, including [28, 18]. Another variation of the artifact-centric approach, called Guard-Stage-Milestone (GSM) [13, 10] is much more declarative and goal-oriented; this is especially useful in supporting collaborative, knowledge-worker-driven BPs. A third variation, EZ-Flow [37], is focused on logical optimization of artifact systems, and the lifecycle model includes a focus on data flow.

Business artifacts are closely related to the notion of “case” in the context of case management systems [31]. Both involve the notion of a conceptual entity that progresses through time, according to some set of guidelines or lifecycle schema, and both taking advantage of a growing set of data accumulated over the case instance lifecycle. Case Management grew out of two related disciplines: (i) Content (or document) Management, which addresses the need in many enterprises to maintain large quantities of heterogeneous documents and files, and (ii) the management of cases in various application domains, such as social work, healthcare, and law, where knowledge workers are given rich flexibility with regards to what activities should be performed and when they should be performed. The Case Management community has recently adopted the GSM approach to provide the basis for the emerging OMG Case Management Model and Notation (CMMN) standard [6, 21].

The artifact-centric and related data-centric approaches have already been applied to support BPs in several application domains, including finance [7], supply chain, retail [4], banking, pharmaceutical research [5], and collaborative work [32, 33]. In many cases, the business managers and subject matter experts involved said that the use of artifacts as the basic modeling primitive gave them a kind of “bird’s eye” view of their operations that they were not obtaining from the traditional activity-flow based approaches, and that it enabled substantially improved communication between the various stakeholders.

There are parallels between the artifact approach to business operations modeling and the Entity Relationship (ER) approach [8] to modeling the data managed in a business. Both are systematic approaches that use a small set of natural and intuitive constructs. Also, artifact specifications are actionable, in the same way that ER diagrams are actionable, i.e. the specification can be used to automatically generate an executable system. However, there is a contrast between how information is typically clustered in artifacts vs. in (relational and ER) database schema design and in document management systems. With database schemas, there is a tendency to break data into fairly small “chunks”: ER-based techniques use separate entity types and their relationships, and relational normal forms break data apart to avoid update anomalies. This is valuable when data is used by a variety of applications. Similarly, document management systems often focus on the company’s literal document types rather than on the single conceptual entity that multiple document types together represent. In contrast, an artifact information model clusters the various kinds of data which correspond to the stages in the dynamic artifact’s lifecycle.

3. EXTENSIONS OF ARTIFACT MODELS

With outsourcing and globalization there is a tremendous and growing need for BPs in different organizations to work together. Classical work on web service interoperation which focuses on a process-centric perspective and supports communication by message passing, has also been applied in the BP context. Generally, there are two approaches to interoperation. In the *orchestrated* approach, a designated “mediator” communicates and coordinates with all participating BPs. This approach is widely used in practice (e.g. via BPEL) but it loses autonomy of participating BPs and does not scale well. The *choreography* approach specifies global behaviors (e.g. in WS-CDL) among participating BPs but otherwise leaves the BPs to operate autonomously and communicate in the peer-to-peer fashion. Notable weaknesses in the current language for interoperation are in representing and using data (e.g. generated by BPs and used in coordination and choreography) and in modeling correlations among participating BP execution instances.

Recently, artifact-centric approaches for both orchestration and choreography have been developed. Because they incorporate data and process in a unified manner, challenges such as on-boarding and modification are simplified. Further, the focus of artifacts on key entities as they progress through cross-enterprise collaborations simplifies the challenge of correlating between messages, process steps, and information that are used by different participants.

[14] introduces a form of orchestration that uses an artifact-centric hub to facilitate the interoperation of multiple enterprise BPs, which are not necessarily artifact-centric. Unlike traditional orchestration schemes, the hub enables the participating BPs to be pro-active, and serves primarily as a shared resource for coordinating activities. Participating BPs can access information about the running artifact instances, can progress those instances along their lifecycles, and can subscribe to events in order to be alerted about significant steps in the progress of artifacts through their lifecycles. Research on artifact-centric interoperation hubs includes work on systems, process mining, verification, and applications [1].

In [30], a choreography language with four distinct features was developed. (1) Each participant type is an artifact BP schema with a selected sub-part of its information model visible to choreography specification. (2) Correlations between participant types and instances are explicitly specified, along with cardinality constraints on correlated instances (e.g., each *Order* instance may correlate with exactly one *Payment* instance and multiple *Vendor* instances). Skolem notations are used to reference correlated participant instances. (3) Messages can include data; data in both messages and artifacts can be used in choreography constraints. Skolem notations are again used to manipulate dependencies among messages and participant instance created by messages. (4) The language is declarative and uses logic rules based on a mix of first-order logic and a set of binary temporal operators from DecSerFlow [36].

An important aspect of BP collaboration is to enable participating organizations to share only part of their information and process. Similar to views in databases, process views aim at providing a mechanism to hide parts of a BP and implicitly, part of the information used in the BP. Most of prior work on process (workflow) views are based on data agnostic models, and consequently working with process views can be intricate because the information portion is implicit. In fact it is a nontrivial and sometimes tricky task to deal with data in developing a tool to support process views. In the artifact-centric context, view definitions specify both what data and what parts of process are to be shared. The data component provides a backbone that simplifies reasoning about information and process sharing. For example, the view mechanism in [38] for artifact-centric BPs formulates consistency criteria for conditions on data in artifacts.

Changes of business policies and operational routines can lead to unavoidable modifications to BP schemas. Currently BPM systems provide basic support in modeling and enactment, but with limited flexibility in handling changes, especially unanticipated and just-in-time changes. Ad hoc changes arise frequently and are witnessed in many business sectors such as banking, clinic trials, and other administrative intensive task management. Various process models, languages, and mechanisms were developed aiming at offering flexibility in BPs. From the language perspective, there are languages [26] that adopt a completely declarative approach to specify process logic without explicit modeling of control flow and data flow as in the procedural BP schemas. From mechanism perspective, FLOWer [2] and ADEPT [29] provide deviation operations such as *undo*, *redo*, and *skip*, and verification techniques to ensure applicability of operations at runtime. From the modeling perspective, most work is based on the traditional process-centric BP mod-

els. In contrast, the artifact-centric approach simplifies managing runtime changes since runtime status information (of enactments) is readily available in artifacts. A hybrid artifact-centric model developed in [37] utilizes the benefits of both declarative and procedural BP schemas. Since the data used in a BP is packaged into an artifact, the runtime status can be easily captured. This provides a solid foundation for managing just-in-time and ad hoc changes. A rule-based language was reported to specify logical conditions when the deviations from specified BP schemas are needed and how the deviations should take place. These rules are managed separately from the process model; updating and changing can be easily applied to the rules without affecting the main workflow model.

Developing robust approaches to managing ad hoc variations in data-centric BP will be important for case management, because the CMMN standard incorporates explicit constructs to allow extensions of BP schemas while case instances are in flight.

4. WORKING WITH LARGE SETS OF PROCESS ENACTMENTS

Business Analytics, also called *Business Intelligence*, is focused on harvesting, mining, and interpreting information from logs and other records of business processes that have been executed. Under the traditional approach, a phase of *Extract-Transform-Load (ETL)* is used to bring the raw data into a unified and meaningful format, followed by a phase of data interpretation and report generation [11]. A central construct for data interpretation is that of *Key Performance Indicators (KPI's)*, which are aggregates that focus on the frequency, distribution and relationships between specified business-relevant events. Some research, e.g., on Business Intelligence Models, develops structures for specifying relationships between KPI's and more coarse-grained strategic goals [23]. A fundamental challenge when using process-centric BPM is that the relationship between process and the data being analyzed is often lost. This can be avoided if business artifacts are used, and also the ETL phase is not needed. For instance, we will discuss a model-driven approach based on business artifacts in which KPI's and metrics are modeled in an integrated manner as part of the business artifact models [19]. The mappings and other low level technical specifications of the monitoring models get generated semi-automatically.

Another area concerned with collections of process enactments is *process mining* [35]. Process mining addresses the problem of mining and analysis of typically large sets of business process enactments (a.k.a. *traces* or *event logs*). This is important in practical settings since it helps to gain insights and understanding of BP enactments, especially in environments of frequent changes. A variety of techniques and applications have been studied. *Inference of BP schemas* from the event logs – also referred to as *discovery of BP schemas* – addresses the challenge of constructing a BP schema which can generate the observed events log. *Conformance Checking* analyzes if observed event logs conform to the known BP specification. Further, extensions such as performance monitoring and techniques for predicting the bottlenecks have been studied. Traditionally, majority of process mining researchers considered the process aspects, employing algorithms such as *alpha algorithm* [35], without putting much emphasis on accompanying data models. Only the recent advances in data-centric business process and case management brought attention to integrated process mining [22, 27, 25] which studies and takes advantage of data as well as process. We focus primarily on the recent work where both data and processes play a significant role since it is very relevant to the databases community and it presents a promising research area.

5. FORMAL MODELS & VERIFICATION

The artifact-centric approach to BP has spawned a very active research program in formal verification in the context of data and process considered together. This area is well motivated, because it is often the case that BP executions must satisfy certain conditions derived from laws, governmental policies, institution policies, contextual requirements, etc. To apply formal verification approaches, it is typical to express these conditions in a temporal logic language. The verification problem is to check if the temporal properties are satisfied by every BP execution. When a BP schema does not include data explicitly, a usual choice of the language for properties is propositional linear temporal logic (PLTL). In this case, checking if every execution of a BP schema would satisfy a given PLTL property can be done by, e.g., model checking. When data is modeled explicitly in the BP schemas, first-order (FO) LTL language and μ -calculus have been used for specifying properties of BP execution. There are two issues, one concerns the modeling of data and the other concerns complexity of the verification problem. In some studies the data in a single artifact instance has been modeled as a set of scalar attributes and possibly with ordered domains and even with arithmetic operations [12, 9], alternatively, [3] permits artifacts whose attributes hold entire relational databases. Both approaches provide insight into the impact of structured data on verification. In general, checking if every possible execution allowed by BP will satisfy a given FO-LTL property is undecidable. The research has discovered restrictions on artifact-centric BP models that yield decidable verification of temporal properties.

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